

Customer Churn Prediction in Telecommunication Industry

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ABSTRACT

Customer churn refers to the situation in which a customer stops using a company's services or switches to another provider. This study focuses on predicting which customers are likely to leave a telecommunications company before the actual churn occurs, using historical customer data and service-related attributes. This prediction is important as it helps organizations reduce potential financial losses and improve customer retention strategies. In this study, machine learning classification models were applied to predict customer churn and compare their performance. The problem was formulated as a binary classification task, where customers are classified as either churn or non-churn. Three models were implemented, namely Decision Tree, Random Forest, and Gradient Boosting. In addition, data preprocessing techniques were applied, including encoding categorical variables, handling missing values, and using cross-validation to ensure robust model evaluation. The results showed that the Gradient Boosting model achieved the highest accuracy (80.08%), while the Random Forest model obtained the highest F1-score (0.63), indicating a better balance between precision and recall. In conclusion, although Gradient Boosting performed better in terms of overall accuracy, Random Forest was found to be more practical for churn prediction, as it is more effective in identifying customers who are likely to leave.

1. INTRODUCTION

The telecommunications industry is currently facing strong competition. Companies are no longer competing only in terms of service quality, but also in terms of their ability to retain customers in a crowded market. Customer churn, which refers to customers leaving their service provider, is one of the major challenges in this sector. High churn rates, which can reach up to 30% in competitive markets, lead to significant financial losses and increased costs of acquiring new customers [1].

Customer churn has become more complex with the rapid growth of data in the telecom industry. Companies now collect large amounts of data from daily activities such as call records, internet usage, and customer service interactions. This has introduced the concept of Big Data, where the volume and variety of data make it difficult to analyze using traditional methods. Understanding customer behavior from this data is essential to identify the reasons behind customer dissatisfaction and churn [2].

To address this challenge, data mining techniques and predictive models are widely used. These methods help analyze large datasets and discover patterns related to customer behavior. In addition, predictive models can identify customers who are likely to leave before it happens. This allows companies to take proactive actions, such as offering personalized services or discounts, which can reduce churn rates and improve customer retention and loyalty [3].

2. LITERATURE REVIEW

Predicting customer churn has been the focus of many studies using various artificial intelligence techniques. This section reviews previous research, focusing on the methods used to process data and the performance of different machine learning and deep learning models. These studies provide a foundation for understanding how to improve prediction accuracy and handle complex customer datasets. This section provides a detailed analysis of the methodologies and findings from these existing works.

2.1 RELATED WORKS

The paper [4] studied the problem of customer churn prediction in the telecom industry, with the goal of identifying the best machine learning model and reducing the number of features without affecting performance.

They first preprocessed the dataset by handling missing and null values, then applied several classification models such as Logistic Regression, SVM, KNN, Decision Tree, Random Forest, Bagging, and XGBoost. After that, they used a correlation matrix to remove less important features, reducing the dataset from 19 to 16 attributes. To improve performance and handle class imbalance, they applied techniques such as SMOTE, ENN, over-sampling, under-sampling, and K-fold cross validation.

The models were evaluated using accuracy, precision, recall, and F1-score. The results showed that Random Forest achieved the best performance, with 90.30% accuracy using all features and 90.90% using the reduced dataset.

This indicates that feature reduction can be applied without negatively affecting performance. The study contributes by combining model comparison, feature selection, and imbalance handling in one framework. However, it is limited to traditional machine learning models and does not explore deep learning or time-based approaches, which are suggested for future work.

In paper [5], the Telco Customer Churn dataset was utilized, and several preprocessing steps were applied, including handling missing values and preparing the data for modeling. They also applied feature selection using correlation to keep only the most important variables. Since the dataset was imbalanced (more non-churn customers than churn customers), they addressed this issue using a data balancing technique called SMOTE (Synthetic Minority Oversampling Technique) combined with ENN (Edited Nearest Neighbours) to improve model performance.

The problem was treated as a classification task, where two machine learning models were used and compared: Decision Tree and Random Forest. The results showed that before applying SMOTE and ENN, the Decision Tree achieved only 78% accuracy, with poor precision, recall, and F1-score for the minority class due to data imbalance. After applying the balancing techniques, the performance improved significantly, reaching about 93.85% accuracy.

For the Random Forest model, it achieved 98.91% accuracy before data balancing and improved further to around 99% after applying SMOTE and ENN, with high precision and recall. The study also highlighted that Random Forest uses a bagging technique (bootstrap aggregating), which combines multiple decision trees trained on different samples of the data to improve overall prediction accuracy and reduce overfitting. Overall, Random Forest outperformed Decision Tree and showed more reliable performance after handling the data imbalance.

Traditional machine learning models often struggle to predict telecommunication customer churn due to complex data relationships and a lack of feature attention mechanisms.

To address this, researchers in [6] proposed "ChurnNet," a novel deep learning architecture integrating a 1D Convolutional Neural Network with Residual, Squeeze and Excitation, and Spatial Attention blocks. The model was evaluated on three imbalanced datasets, including the IBM Telco dataset, utilizing 10-fold cross-validation and various data balancing techniques. ChurnNet outperformed existing state-of-the-art methods; specifically, pairing it with the SMOTEEN balancing technique and ADAM optimizer yielded peak accuracies of 95.59%, 96.94%, and 97.52% across the respective datasets. However, the authors note that future work should address the model's "black box" nature by integrating Explainable AI (XAI) and explore decentralized prediction frameworks via federated learning.

Also, the study [7] proposed a hybrid deep learning model that combines Bidirectional Long Short-Term Memory (BiLSTM) and Convolutional Neural Networks (CNN). The model was evaluated using a publicly available Telco dataset from Kaggle, which contains 7,033 customers and 20 features, including demographic and service-related attributes such as age, monthly charges, and contract type. The dataset was divided into 80% for training and 20% for testing.

Before applying the model, several preprocessing steps were performed. These included data cleaning, converting categorical (text) data into numerical form, handling missing

values, and normalizing the data to improve model performance.

The proposed model operates in two main stages. First, the BiLSTM layer processes the data in both forward and backward directions, allowing it to capture temporal and sequential patterns in customer behavior over time. This helps in understanding how customer behavior evolves and whether it indicates a likelihood of churn. Second, the CNN layer takes the output from the BiLSTM and extracts important features that enhance the classification process. Finally, a sigmoid function is used for classification, where a value greater than 0.5 indicates that the customer is likely to churn, while a value less than or equal to 0.5 indicates that the customer will stay.

The experimental results demonstrated that the proposed BiLSTM-CNN model achieved an accuracy of 81%, with a precision of 66%, recall of 64%, and F1-score of 65%. These results outperformed several traditional machine learning models, including KNN (71%), Decision Tree (78%), SVM (76%), and XGBoost (79%). Additionally, the hybrid model showed better performance compared to individual deep learning models such as CNN and RNN, due to its ability to combine sequential learning and feature extraction.

Despite these promising results, the study [8] has some limitations. It used a relatively limited number of features (only 20), and it evaluated only one hybrid model without exploring other possible architectures. Expanding the dataset and experimenting with different models could further improve the results.

2.2 SUMMARY

Previous studies applied machine learning and deep learning models for customer churn prediction, mainly treating it as a classification problem. Common approaches included ensemble methods such as Random Forest and deep learning models, along with preprocessing steps like feature selection and handling class imbalance.

Results showed that Random Forest often performed strongly, while deep learning models achieved higher accuracy in some cases. It was also observed that addressing class imbalance improves model performance significantly.

However, some studies lacked clear comparisons between machine learning and deep learning models, and many focused mainly on accuracy rather than F1-score, which is more suitable for imbalanced datasets. In addition, class imbalance was not always properly handled. Therefore, this study compares multiple models while emphasizing F1-score and explicitly considering class imbalance to achieve more reliable churn prediction.

3. DATASET DESCRIPTION

The dataset for this study is the Telecommunications Industry Sample Data, obtained from IBM Cognos Analytics. It is a set of sample data that provides information about a fictional telco company. The primary data is stored in a file named

"CustomerChurn.csv", which contains 7,043 customer records and 21 attributes. The "Churn" column is our target.

The following table provides a detailed description of each attribute found in the "CustomerChurn.csv" dataset.

Attribute	Description	Data Type
LoyaltyID	A unique ID assigned to each customer for the loyalty program.	Numerical
Customer ID	Unique identifier for each customer.	Object / Categorical
Senior Citizen	Whether the customer is a senior citizen or not (1, 0)	Binary
Partner	Whether the customer has a partner or not (Yes, No)	Binary
Dependents	Whether the customer has dependents or not (Yes, No)	Binary
Tenure	Number of months the customer has stayed with the company	Numerical
Phone Service	Whether the customer has a phone service or not (Yes, No)	Binary
Multiple Lines	Whether the customer has multiple lines or not (Yes, No, No phone service)	Categorical
Internet Service	Customer's internet service provider (DSL, Fiber optic, No)	Categorical
Online Security	Whether the customer has online security or not (Yes, No, No internet service)	Categorical
Online Backup	Whether the customer has online backup or not (Yes, No, No internet service)	Categorical
Device Protection	Whether the customer has device protection or not (Yes, No, No internet service)	Categorical
Tech Support	Whether the customer has tech support or not (Yes, No, No internet service)	Categorical
Streaming TV	Whether the customer uses internet to stream TV.	Categorical
Streaming Movies	Whether the customer uses internet to stream movies.	Categorical
Contract	The contract term of the customer (Month-to-month, One year, Two year)	Categorical
Paperless Billing	Whether the customer has paperless billing or not (Yes, No)	Binary
Payment Method	The customer's payment method (Electronic check,	Categorical

	Mailed check, Bank transfer (automatic), Credit card (automatic))	
Monthly Charges	The amount charged to the customer monthly	Numerical
Total Charges	The total amount charged to the customer	Numerical
Churn	Whether the customer churned or not (Yes or No)	Binary

Table 1: Dataset Attributes Description

4. DATA MINING TECHNIQUES

This study uses classification techniques to predict customer churn. Classification is suitable for this problem because the output variable is binary, where customers are classified as either churn or non-churn. Formally, let the historical dataset be defined as $D = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^d$ represents the d -dimensional feature vector of the i -th customer (e.g., demographic and service-related attributes), and $y_i \in \{0,1\}$ is the binary target variable, with 1 representing a churned customer and 0 representing a retained customer.

Machine learning classification models learn patterns from historical data to approximate a mapping function $f : X \rightarrow Y$. The objective is to minimize the prediction error so that the learned function can be generalized to make predictions on new data, and in this study, to predict future customer behavior accurately.

5. PROPOSED ALGORITHMS

We evaluated three machine learning algorithms: Decision Tree [8], Random Forest [9], and Gradient Boosting [10]. These tree-based and ensemble learning algorithms were selected because they are widely used in classification problems and can handle structured data effectively and capture complex relationships between features.

5.1 DECISION TREE

A Decision Tree algorithm [8] partitions the feature space into distinct regions by continuously splitting the data based on specific feature values. The optimal splits are determined by minimizing an impurity measure at each node, commonly the Gini Impurity. For a binary classification problem, the Gini Impurity at a given node is calculated as:

$$Gini(p) = 1 - \sum_{i=1}^c p_i^2$$

Where C is the total number of classes (in this case, 2), and p_i^2 is the probability of a sample belonging to class i at that specific node.

5.2 RANDOM FOREST

Ensemble models such as Random Forest [9] are expected to perform better due to their ability to combine multiple learners. Random Forest builds an ensemble of independent decision trees using a technique called bootstrap aggregating (bagging). By training multiple deep trees on different subsets of the data, the model reduces variance and prevents overfitting. The final prediction is determined by a majority vote across all the individual trees:

$$\hat{y} = \underset{c}{\operatorname{argmax}} \sum_{b=1}^B I(\hat{y}_b = c)$$

Where B is the total number of decision trees in the forest, $\hat{y}_b$ is the class prediction of the b -th tree, and $I(\cdot)$ is the indicator function that outputs 1 if the condition is met and 0 otherwise.

5.3 GRADIENT BOOSTING

Gradient Boosting [10] is another ensemble technique, but unlike Random Forest, it operates sequentially. It builds a series of weak learners (typically shallow decision trees) where each new tree attempts to correct the residual errors made by the combined ensemble of previous trees. The model updates additively and minimizes a loss function using gradient descent, defined by the following step:

$$F_m(x) = F_{m-1}(x) + \nu h_m(x)$$

Where $F_m(x)$ is the model at stage m , $h_m(x)$ is the weak learner fitted to the pseudo-residuals of the previous stage, and $\nu \in (0,1]$ is the learning rate that shrinks the contribution of each tree to prevent overfitting.

6. METHODOLOGY

This section outlines the systematic approach used to develop and evaluate the customer churn prediction models. The research workflow is divided into four key stages: analyzing attribute correlations, preprocessing the dataset, configuring the experimental setup for the chosen algorithms, and evaluating the final model performance.

6.1 ATTRIBUTE CORRELATION

A correlation analysis was conducted to understand the linear relationship between the features and the target variable (Churn). The results indicated that the highest positive correlation with customer churn was *Monthly Charges* (0.193). Conversely, the highest negative correlation was observed with the *Contract* attribute (-0.397), suggesting that longer-term contracts are strongly associated with lower churn rates.

6.2 DATA PREPROCESSING

Data preprocessing was performed to prepare and ensure the dataset was suitable for the machine learning models.

1. First, irrelevant features such as *Customer ID* and *LoyaltyID* were dropped, as they are unique identifiers that do not contribute to predictive patterns.
2. Second, the *Total Charges* column was converted to a numeric format, and any missing (null) values were

handled by imputing them with the median value of the column to prevent data loss.

3. Finally, because machine learning algorithms require numerical input, all categorical variables (such as contract type, and payment method) were converted into numerical values using Label Encoding.

6.3 EXPERIMENTAL SETUP

To ensure robust model evaluation and mitigate the effects of the class imbalance (5,174 retained vs. 1,869 churned), a Stratified 5-Fold Cross-Validation approach was implemented. This technique ensures that each fold maintains the same proportion of churned to non-churned customers as the original dataset.

Three classifiers were configured with specific hyperparameters to optimize their performance:

- **Decision Tree:** Configured with a maximum depth of 6, a minimum of 10 samples required to split an internal node, and a minimum of 5 samples required to be at a leaf node.
- **Random Forest:** Built using an ensemble of 300 decision trees (`n_estimators=300`) with a maximum depth of 10 and a minimum sample split of 5. Additionally, the `class_weight` parameter was set to "balanced" to automatically adjust weights inversely proportional to class frequencies, directly addressing the dataset imbalance.
- **Gradient Boosting:** Configured with 150 estimators, a learning rate of 0.05 to scale the contribution of each tree, and a maximum depth of 4.

6.4 RESULTS AND EVALUATION

The Gradient Boosting model achieved the highest overall performance, with an accuracy of 80.08%, an F1-Score of 0.58, and an AUC of 0.85. As shown in the Aggregated Confusion Matrix (Figure 1), the model correctly classifies the majority of customers, particularly those who do not churn. The ROC Curve (Figure 2) also demonstrates strong classification ability across different thresholds. This high performance is due to the model's ability to learn from previous errors and improve iteratively. However, due to class imbalance, the model focuses more on overall accuracy, which makes it less effective in capturing all churn cases compared to other models, resulting in a lower F1-Score.

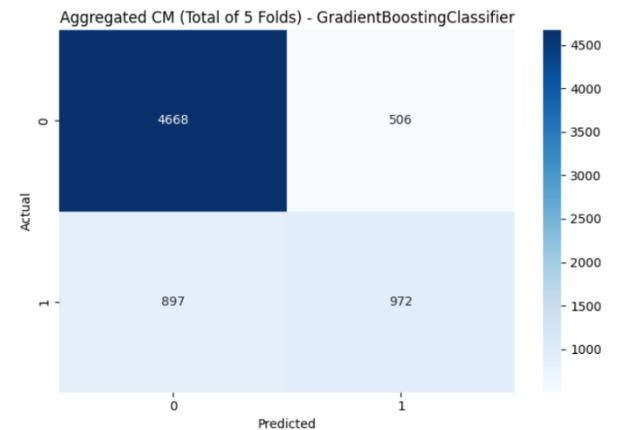


Figure 1: Gradient Boosting Confusion Matrix

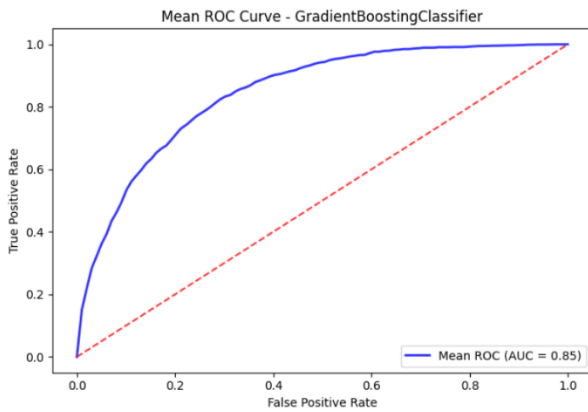


Figure 2: Gradient Boosting ROC curve

The Random Forest model achieved an accuracy of 77.65%, an F1-Score of 0.63, and an AUC of 0.84. According to the Aggregated Confusion Matrix (Figure 3), it correctly identifies a higher number of churn cases compared to the other models. The ROC Curve (Figure 4) further supports its strong performance. Although its accuracy is slightly lower, the higher F1-Score indicates a better balance between precision and recall. This makes Random Forest the most effective model for detecting customers who are likely to churn, which is important for retention strategies.

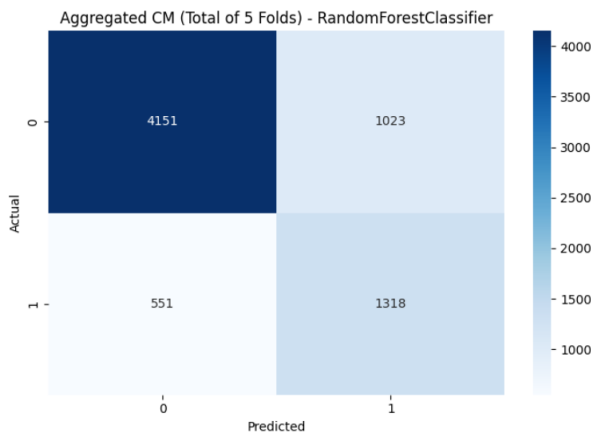


Figure 3: Random Forest Confusion Matrix

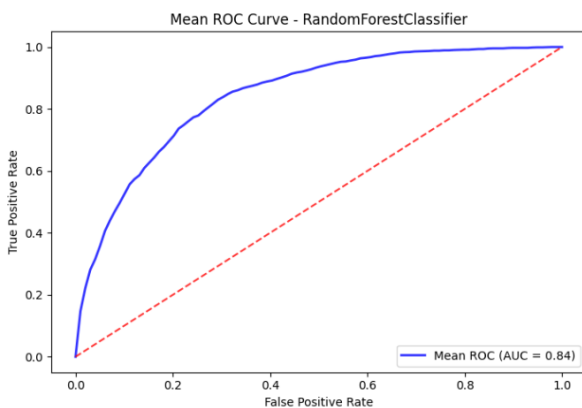


Figure 4: Random Forest ROC curve

The Decision Tree model achieved an accuracy of 78.23%, an F1-Score of 0.54, and an AUC of 0.82. As illustrated in the Aggregated Confusion Matrix (Figure 5), the model performs reasonably well but misclassifies more churn cases compared to the other models. The ROC Curve (Figure 6) shows moderate performance. While it does not outperform ensemble models, it is simple and easy to interpret. This makes it a useful baseline model and highlights the impact of key features such as contract type and monthly charges on customer churn.

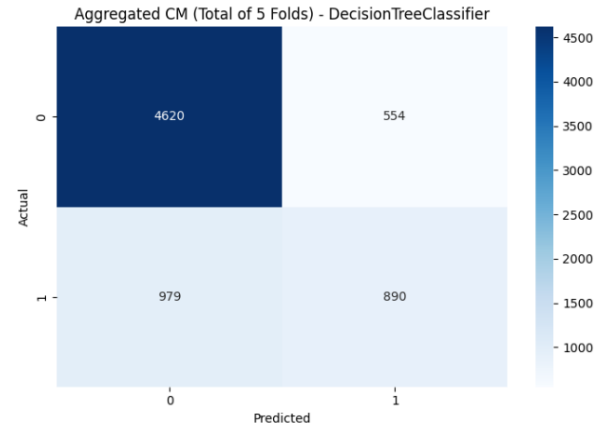


Figure 5: Decision Tree Confusion Matrix

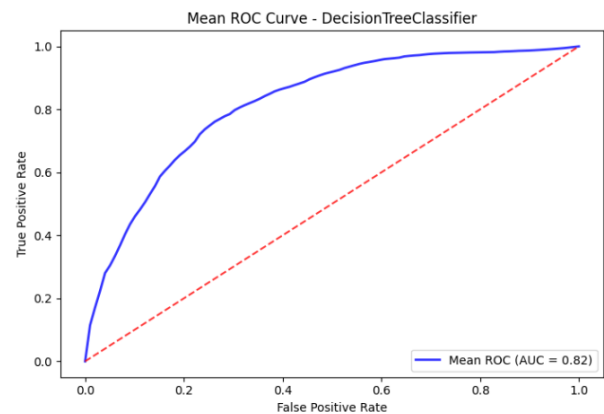


Figure 6: Decision Tree ROC curve

In summary, while the Gradient Boosting model achieved the highest overall accuracy (80.08%) and AUC (0.85), the Random Forest model is more suitable for churn reduction. This is because it achieved the highest F1-Score (0.63), indicating a better balance in identifying customers who are likely to leave. Since the main goal is to detect at-risk customers, Random Forest provides more reliable results for targeting retention strategies.

7. CONCLUSION

The main objective of this study was to predict customer churn using machine learning models. Three models were implemented, namely Decision Tree, Gradient Boosting, and Random Forest, along with data preprocessing and cross-validation to ensure reliable evaluation.

Although the Gradient Boosting model achieved the highest accuracy, this does not necessarily mean it is the most effective

model in practice. In churn prediction problems, the ability to correctly identify customers who are likely to leave is more important than overall accuracy. For this reason, the Random Forest model is considered more suitable, as it achieved the highest F1-score and provided a better balance between precision and recall. This indicates that Random Forest is more effective in detecting potential churn customers, making it more useful for real-world applications such as customer retention strategies.

However, the performance of the models was influenced by the imbalanced nature of the dataset, which may limit the ability to detect all churn cases accurately. Therefore, future work could focus on applying data balancing techniques and exploring additional models to further improve performance.

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